

Problem-Solution fit canvas 2.0 – AI-Driven Healthcare Anomaly Detection System

Purpose / Vision:
Real-time Healthcare Monitoring & Early Risk Alerts

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Hospitals & Healthcare Providers - ICU & Critical Care Units - Doctors & Healthcare Administrators - Patient Monitoring Centers - Health Tech Companies & Researchers	6. CUSTOMER CONSTRAINTS CC - High cost of continuous monitoring systems - Lack of real-time anomaly detection - Limited access to accurate & timely alerts - Data privacy & security concerns - Complex integration with existing systems	5. AVAILABLE SOLUTIONS AS - Traditional vital sign monitors (static thresholds) - Manual patient observation - Post-incident health records analysis - Basic alert systems (rule-based) - Wearable devices with limited intelligence	Explore AS, differentiate
Focus on J&P, fit into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Monitor patient vitals in real-time - Detect health anomalies early - Reduce delay in critical care - Automate health risk alerts - Improve patient safety & outcomes	9. PROBLEM ROOT CAUSE RC - Delayed detection of health issues - Manual monitoring errors - No real-time analytics on vitals - Overload on healthcare staff - Lack of integrated alert system	7. BEHAVIOUR BE - Doctors rely on dashboards & alerts - Admins monitor critical patients 24/7 - Quick response to high-risk alerts - Trust AI-based decision support - Prefer real-time + email notifications	Focus on J&P, fit into BE, understand RC
Identify using TR & EN	3. TRIGGERS TR - Abnormal vital signs detected - Patient's health deterioration - Critical event/ Emergency - Scheduled monitoring alerts - Doctor/ Admin login activity 4. EMOTIONS: BEFORE / AFTER EM Before: Stress, uncertainty, delayed action After: Confidence, safety, quick response - Peace of mind, improved efficiency	10. YOUR SOLUTION SL - Ai-powered real-time health monitoring system - Anomaly detection using Autoencoder & Isolation Forest - Severity alerts (LOW, MEDIUM, HIGH) - Interactive dashboard + email notifications - Secure database & scalable streaming pipeline	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE - Web Dashboard (Flask) - Email Alerts & Notifications - Admin/Doctor Login Portal 8.2 OFFLINE - Hospital Systems integration - Reports & Data Storage (PostgreSQL) - Emergency Alerts (Email/SMS)	Enriched online & offline CH of BE